

SurgeNet-ACT: A Foundation-Model Framework and Benchmark for Autonomous dVRK Suturing

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Abstract—Autonomous suturing on the da Vinci Research Kit (dVRK) is learned from a few dozen teleoperated demonstrations and must keep working on the same robot across days. We present SurgeNet-ACT, a framework that encodes the camera views with SurgeNetXL, a frozen surgical foundation model, and predicts action chunks with an ACT policy as displacements and rotations in the tool frame of the measured pose, with recent measured poses as temporal context. We evaluate on the public SutureBot benchmark and on the SurgLab Benchmark, 84 bimanual demonstrations of needle pickup, needle throw and knot tying from three sessions. The frozen encoder lowers position error on every core phase of SutureBot’s held-out tissue and cuts offset-corrected error on our robot from 4.5 / 3.8 to about 3.2 mm; LoRA, partial fine-tuning and cross-dataset pre-training add nothing. Tool-frame actions make the policy insensitive to the calibration change between sessions, cutting raw error on an unseen session from 4.6 to 1.06 mm and rotation error from 21 to 3.1 degrees. The final policy reaches 0.78 / 0.79 mm and 3.0 degrees on two unseen sessions at 30 Hz, and a per-demonstration protocol shows where offline evaluation stops ranking policies. Dataset and code are available at [URL].

Index Terms—surgical robotics, imitation learning, autonomous suturing, foundation models, da Vinci Research Kit

I. INTRODUCTION

Suturing is the most repetitive fine-motor task in minimally invasive surgery and the standard test of surgical robot learning on the da Vinci Research Kit (dVRK) [1]–[4]. Each stitch has three phases, needle pickup, needle throw and knot tying, and each needs both arms to place a curved needle to within a millimetre. It is hard for a learned policy: both arms act at once on a needle that travels a few millimetres per chunk, the scene is seen only through cameras, and a laboratory has a few dozen demonstrations to learn from. A policy is useful to that laboratory only if it can be trained on its own robot, from its own demonstrations, and kept in use across days.

Recent systems show that suturing can be learned from demonstrations, but none is evaluated on a later recording session of the same robot. The Surgical Robot Transformer (SRT) [3] showed that action-chunking policies [5] trained from teleoperation perform suturing sub-tasks on the dVRK; its successor SRT-H [6] completed autonomous cholecystectomy on unseen ex vivo gallbladders under language-conditioned corrections; SutureBot [4] released 1,890 suturing demonstrations with a multi-task ACT baseline. All three learn their visual features from the task data with a convolutional

encoder, regress absolute end-effector poses, and are evaluated within one recording setup. On a research dVRK the camera-to-robot calibration changes between sessions, and a policy that regresses absolute poses inherits the offset in full, 2 to 26 mm on our robot (Sec. IV-E).

The gap this paper closes is a recipe that a laboratory can train from a few dozen demonstrations, that survives the next recording session of its own robot, and that comes with the data to reproduce it.

We close it with SurgeNet-ACT (Fig. 1). The camera views are encoded by SurgeNetXL, a surgical foundation model that stays frozen, into patch tokens shared across views. An ACT policy reads the tokens, the phase command and the jaw angles and predicts each action chunk as displacements and rotations in the tool frame of the measured pose. The recent measured poses, rewritten in that frame, serve as temporal context. Everything else is the ACT recipe of SutureBot, so each stage is measured against the recipe it replaces.

We evaluate on two benchmarks. The public SutureBot Benchmark, 1,879 training demonstrations recorded on a different dVRK and evaluated on a held-out tissue sample, tests the encoder and the policy head at scale. Our SurgLab Benchmark, 84 bimanual demonstrations of needle pickup, needle throw and knot tying recorded in three sessions, tests the policy on sessions never seen in training. Our contributions are:

- **Framework.** SurgeNet-ACT combines a frozen surgical foundation encoder, an ACT policy with tool-frame action chunks and temporal context from measured poses, and runs at 30 Hz on one GPU (Sec. III).
- **Foundation encoder evaluation.** Frozen SurgeNetXL beats the task-trained convolutional encoders of the SutureBot recipe on both benchmarks, and LoRA, partial fine-tuning and cross-dataset pre-training do not improve it (Tables I and III).
- **Two-benchmark results.** On SutureBot the CVAE head of ACT ties VQ-BeT and beats diffusion on offline L1; on SurgLab tool-frame actions and temporal context reach 0.78 / 0.79 mm and 3.0 degrees on two unseen sessions, and a hold-pose floor and direction diagnostic show where offline error stops ranking policies (Tables VII and VIII).
- **Dataset and code.** We release the SurgLab Benchmark with its splits and protocol, and the training and deployment code (Sec. IV-B).

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Sec. II positions the work, Sec. III describes the framework and the protocol, Sec. IV the benchmarks and results, and Sec. V what they do and do not show.

II. RELATED WORK

Autonomous suturing. STAR [2] performed autonomous laparoscopic intestinal anastomosis with sutures planned from fiducial markers; since then, learning from demonstration dominates. SRT [3] trains action-chunking policies for suturing sub-tasks on the dVRK and compares camera-centric, tool-centric and hybrid-relative action formulations, the last of which gave the best closed-loop success. SutureBot [4] released a 1,890-demonstration benchmark with a multi-task ACT baseline that predicts absolute targets. Both learn their visual features from the task data and evaluate within one recording setup. We keep SRT’s tool-centric formulation and SutureBot’s head, replace the encoder with a frozen surgical foundation model, and evaluate on sessions the policy never saw.

Language- and vision-conditioned surgical policies. SRT-H [6] adds a language-conditioned high-level policy that corrects a low-level action-chunking policy and completes autonomous cholecystectomy on unseen ex vivo gallbladders; generalist vision-language-action models such as GR00T N1 [7], a vision-language module coupled to a diffusion transformer, scale the policy across robots and tasks; SutureBot evaluates it on its benchmark. SutureBot’s baseline is conditioned the same way, on a phase command and on correction commands for its recovery variants. All of these change the policy; we change what it sees and what it predicts, independent of policy size or language interface, and keep the language input to the phase command.

Surgical foundation models and visual backbones. Self-supervised vision transformers such as DINOv2 and DINOv3 [8]–[10] provide dense patch features that transfer across domains without labels. SurgeNet [11] showed that pre-training on surgical video improves surgical downstream tasks over natural-image pre-training. SurgeNetXL [12] scales the recipe to 4.7 million frames and, across six datasets, beats the best ImageNet-based variants by 14.4, 4.0 and 1.6 percent on semantic segmentation, phase recognition and critical-view-of-safety classification; its released checkpoints cover DINOv1, DINOv2 and DINOv3 objectives at ViT-S, B and L sizes. All of these evaluations are perception tasks. We use the SurgeNetXL DINOv3 ViT-B/16 checkpoint frozen inside a control policy, measure it against the task-trained encoders of the SutureBot recipe on two benchmarks, and quantify what LoRA [13], partial fine-tuning and cross-dataset pre-training add on top: nothing at our data scale.

Imitation learning with action chunking. ACT [5] predicts a chunk of future actions from a CVAE latent in one transformer pass and smooths overlapping chunks by temporal ensembling. Diffusion Policy [14] denoises an action sequence conditioned on the observation, and VQ-BeT [15] quantises actions into discrete codes and predicts them with a transformer; both represent multimodal action distributions.

On surgical data, SRT [3] and SutureBot [4] both adopt ACT with task-trained convolutional encoders and report closed-loop success under one recording setup. We keep ACT’s chunking, compare the three heads on the same frozen features (Sec. IV-D), and evaluate at the executed horizon, the part of a chunk a deployment sends to the robot, on sessions the policy never saw.

Surgical datasets and benchmarks. JIGSAWS [16] pairs dVRK kinematics and video of suturing with gesture labels and benchmarks for gesture segmentation and recognition, and SAR-RARP50 [17] provides 50 suturing segments from real prostatectomies with action and tool labels but no kinematics; neither is meant for closed-loop control. SutureBot [4] pairs 1,890 dVRK demonstrations with kinematics, three views and clicked needle-throw goals under one calibration and reserves one tissue sample for evaluation; SRT [3] trains on its own dVRK recordings. Our SurgLab Benchmark (Sec. IV-B) is smaller but records three sessions, measured and commanded poses, four views, the clicked needle-throw goals and a sub-task annotation of 289 clips, which is what training and testing a policy across sessions of one robot needs.

III. METHOD

A. Task and Problem Formulation

The task is bimanual suturing on the dVRK in three phases: needle pickup, needle throw and knot tying, each executed by two patient-side manipulators (PSM1, PSM2). At step t the policy observes C camera images ($C = 4$ on our robot, 3 on SutureBot), the two jaw angles and a text command naming the phase. The state it acts from is the measured pose ($\mathbf{p}_0, \mathbf{R}_0$) of each arm, which serves only as the reference frame of the outputs. The output is an action chunk of T steps with 20 numbers per step: for each arm, 3 translation, 6 rotation and 1 jaw value. A deployment executes the first H steps of each chunk before the next chunk replaces it. Success is measured offline, as the position and rotation error of the executed part of the chunk against the demonstrated setpoints on sessions never seen in training (Sec. III-G).

B. Framework Overview

SurgeNet-ACT has four stages (Fig. 1). (1) A frozen SurgeNetXL encoder turns each camera view into a grid of patch tokens. (2) The tokens, the command, the jaw angles, a CVAE latent and the temporal-context tokens form the memory of an ACT transformer encoder. (3) An ACT decoder reads out the chunk in the tool frame of the measured pose. (4) The chunk is composed with the measured pose into absolute setpoints for the dVRK. The measured pose enters only at stage (4). Secs. III-C to III-E describe the stages, III-F the loss and III-G the protocol.

C. SurgeNet Backbone

SurgeNetXL [12] is a family of vision transformers pre-trained self-supervised on 4.7 million surgical video frames; we use its DINOv3 ViT-B/16 checkpoint. Each view is resized to 240×320 and encoded, frozen and shared across views, into

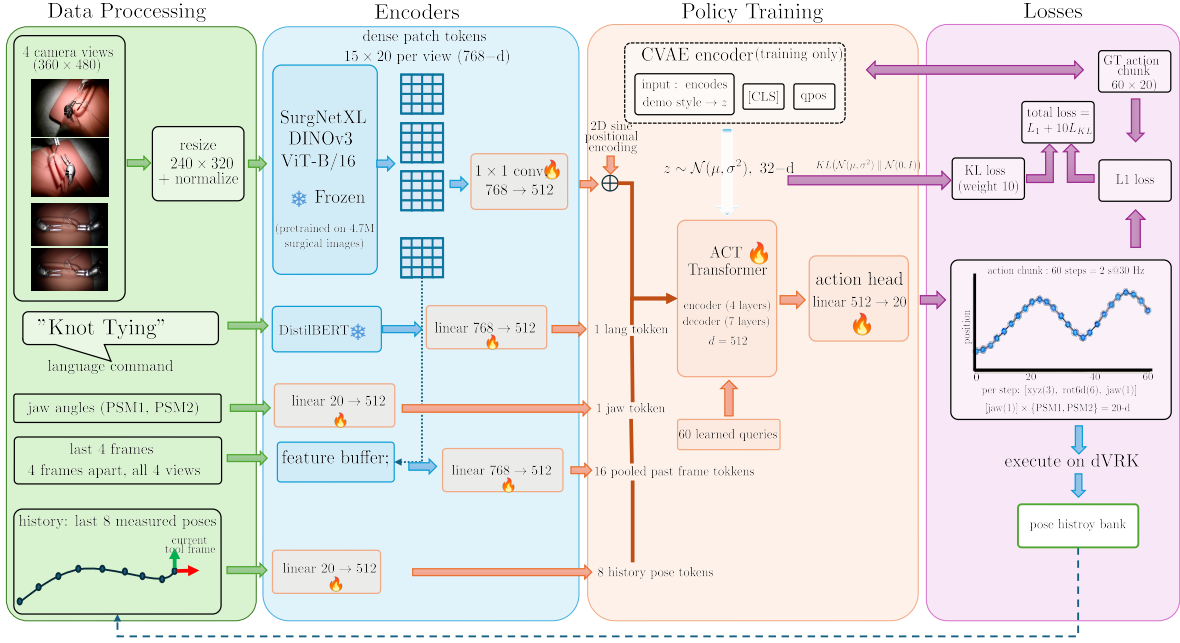


Fig. 1. SurgeNet-ACT framework. Left to right: the four camera views, the phase command, the jaw angles and the recent frames and measured poses are encoded (frozen SurgeNetXL and DistilBERT in blue, trained layers in orange); an ACT transformer with a CVAE latent decodes a chunk of tool-frame actions; the loss (2) trains it and the chunk is executed on the dVRK.

a 15×20 grid of 768-d patch tokens. A trained 1×1 projection brings the 300 tokens per view to $d = 512$; each token receives a 2D sine position encoding and, on our robot, a learned per-view embedding that tells the views apart. Because the encoder is frozen, its tokens are computed once and cached, and most runs train from the cache; we also test three ways of adapting it: LoRA [13] on the last four blocks, fine-tuning the last two blocks, and initialising the head from a policy pre-trained on SutureBot (Sec. IV-E).

D. ACT Policy

The policy is the CVAE encoder–decoder of ACT [5] as used by SutureBot. Three inputs reach d through their own trained linear layers: the phase command, embedded once by a frozen DistilBERT [18]; the proprioception vector, which carries only the two jaw angles; and the 32-d CVAE latent $\mathbf{z}$, inferred from the ground-truth chunk during training and set to zero at deployment. With the image tokens and the temporal-context tokens of Sec. III-E these form the memory of a 4-layer transformer encoder; a 7-layer decoder with T learned queries, one per chunk step, reads out the chunk through a linear head, $512 \rightarrow 20$. ACT can smooth overlapping chunks at inference by temporal ensembling, an exponentially weighted average of the predictions made for the same step; the SutureBot recipe keeps this option, whereas our deployment wrapper executes the first H steps of each chunk without it, as does every offline number in this paper.

E. Tool-Frame Actions and Temporal Context

Let $(\mathbf{p}_0, \mathbf{R}_0)$ be the measured pose of an arm at the start of a chunk and $(\mathbf{p}_t, \mathbf{R}_t)$ the demonstrated setpoint at chunk step

t . The policy predicts, for each arm and step,

$$\mathbf{d}_t = \mathbf{R}_0^\top (\mathbf{p}_t - \mathbf{p}_0), \quad \mathbf{R}_t^{\text{rel}} = \mathbf{R}_0^\top \mathbf{R}_t, \quad \dot{j}_t, \quad (1)$$

where $\dot{j}_t$ is the absolute jaw angle and $\mathbf{R}_t^{\text{rel}}$ is encoded by its first two columns [19]. The targets describe where the tool goes as seen from where it is, so a rigid transform applied to every pose, such as a per-session calibration error, cancels in both terms, whereas a camera-frame translation delta cancels only its translation part. At deployment the predicted deltas are composed with the measured pose, $\mathbf{p}_t = \mathbf{p}_0 + \mathbf{R}_0 \mathbf{d}_t$, $\mathbf{R}_t = \mathbf{R}_0 \mathbf{R}_t^{\text{rel}}$, after Gram–Schmidt orthonormalisation of the predicted rotation. Positions and jaw angles are standardised with the mean and standard deviation of the relative targets over the training set; the six rotation entries are left unscaled. For the needle-throw phases the benchmark’s goal conditioning is kept. The two clicked points are blended into the left endoscope image as a red and a green disc of radius 10 pixels at 50 percent opacity, in training and at deployment alike.

Temporal context gives the policy the recent past from measurements, never from its own outputs, so it cannot learn to extrapolate its action history and ignore vision. *Pose history.* The $K = 8$ measured poses at 4, 8, $\dots$, 32 frames back are each rewritten in the current tool frame with (1). This gives 20 numbers per step for both arms, projected $20 \rightarrow 512$ by one shared linear layer plus a learned slot embedding. *Image history.* Each time SurgeNetXL encodes a frame, the 300 tokens of each view are mean-pooled into one 768-d vector and stored; the vectors of the $F = 4$ frames at 4 to 16 frames back, 16 tokens, are projected $768 \rightarrow 512$ with a slot embedding, so no past image is re-encoded. In 30 percent of training samples

all history tokens are replaced by one learned null token, so the policy also works from an empty buffer.

F. Training Objective

The loss trains the head to reproduce the demonstrated chunk while keeping the CVAE latent close to its prior. Let $\hat{\mathbf{a}}_t, \mathbf{a}_t \in \mathbb{R}^{20}$ be the predicted and demonstrated action at chunk step t (both arms, standardised) and $\mathcal{P}$ the eight position and jaw entries. Let $\hat{\mathbf{R}}_{t,k}, \mathbf{R}_{t,k}$ be the rotation matrices of arm k recovered from the predicted and demonstrated 6D vectors by Gram–Schmidt orthonormalisation, and $v_t \in \{0, 1\}$ the mask that is zero for chunk steps past the end of the demonstration. Then

$$\mathcal{L} = \frac{1}{\sum_t v_t} \sum_t v_t \left(\frac{\|\hat{\mathbf{a}}_t - \mathbf{a}_t\|_{1,\mathcal{P}}}{|\mathcal{P}|} + \frac{1}{2} \sum_k \frac{\|\hat{\mathbf{R}}_{t,k} - \mathbf{R}_{t,k}\|_1}{9} \right) + \beta \text{KL}(q(\mathbf{z} | \mathbf{a}) \| \mathcal{N}(0, I)), \quad (2)$$

with $\beta = 10$. The rotation term compares orthonormalised matrices, so the raw 6D outputs are constrained only up to the scale the orthonormalisation removes; Sec. IV-E also tests a variant that weights early chunk steps up to twice the late ones, doubles the jaw term and adds $1 - \cos$ between predicted and demonstrated translation deltas. The same orthonormalised matrix serves as the validation metric, which removes a scale drift of the raw 6D outputs; unlike the reference implementation, we validate in deployment mode (Sec. III-G), with the latent set to zero. We train with AdamW, cosine decay after a warm-up of one percent of the epochs, an exponential moving average of the trainable weights (decay 0.999) for validation and deployment, and bf16 autocast. The learning rate is 1.3×10^{-4} at batch 56 for image-based runs and 1.6×10^{-4} at batch 96 for runs on cached backbone tokens. Image-based runs use the SutureBot augmentation pipeline, whose large random image shifts were the strongest regulariser; runs on cached tokens instead drop each camera with probability 0.1 (at most two) and each token with probability 0.1. The SutureBot policies (Sec. IV-D) use the benchmark’s own loss, L1 on all 20 dimensions with KL weight 10, no per-view embedding, batch 76 and learning rate 1.48×10^{-4} .

G. Evaluation Protocol

On our robot, for each validation demonstration we draw 40 start frames with a fixed random seed and run the policy in deployment mode, latent set to zero and history built from measured poses only. The predicted chunk is compared with the demonstrated setpoints over the first H steps, 240 chunks per policy and horizon, at the executed horizon $H = 20$ (0.67 s) and over the full chunk. We report per arm the mean Euclidean position error, the mean geodesic rotation error, and the *offset-corrected* error obtained after subtracting each demonstration’s mean error vector, whose norm is reported as the offset; the subtraction separates a constant per-session offset from the policy’s own error. Relative-action predictions are composed with the ground-truth pose at the chunk start so that all policies are compared in the same frame. For those

we also report, on the same chunks, the error of holding the current pose for H steps, the floor of this metric. A direction diagnostic completes the protocol: the cosine between predicted and demonstrated tool-frame translation over the horizon and the ratio of their magnitudes. Mean $\pm$ std is over training seeds on identical chunks; rows without a spread are single runs, and differences smaller than nearby spreads are not rankings. Checkpoints are selected by the lowest deployment-mode validation L1 over the 20 action dimensions, with the rotation slots replaced by their orthonormalised 6D encoding (Sec. III-F). On SutureBot we keep the benchmark’s own metric, per-axis L1 of the predicted chunk over every frame of the evaluation demonstrations.

IV. EXPERIMENTS

A. SutureBot Benchmark

The public SutureBot dataset [4] was recorded on a different dVRK with three views (endoscope and two wrist cameras) and one fixed camera-to-robot calibration. It is organised by tissue sample, with three phases and a perturbed recovery variant of each. Tissue samples 1 to 9 hold 1,879 demonstrations (657,605 steps at 30 Hz); tissue sample 10 holds the reserved evaluation set of 12 demonstrations, two per phase and recovery variant (4,693 steps). We train on tissues 1 to 9 and evaluate on every frame of tissue 10, an out-of-distribution evaluation: the phantom, its placement and lighting never appear in training. SutureBot’s own evaluation is closed-loop success on its robot, which we cannot reproduce; we report the per-phase offline chunk error on this set instead. Actions are absolute poses in the robot frame, 60-step chunks.

B. SurgLab Benchmark Dataset

Platform and sensors. Our dVRK has two patient-side manipulators (PSM1, PSM2), a stereo endoscope (960×540) and one wrist camera per arm (640×480), all logged at 30 Hz, plus an RGB-D camera the policies do not use. Each frame stores, for each arm, the measured pose, the commanded setpoint and the jaw angle, plus the joint states of the arms, the endoscope and the setup joints. Poses are re-expressed in the left endoscope camera frame with a per-arm extrinsic transform from a marker-based calibration; the calibration is redone when the endoscope or setup joints move.

Demonstrations. Four operators teleoperated 28 trials of the three phases, giving 84 demonstrations (50,502 frames, 28.0 minutes). 78 were recorded on the first day in one session and 6 on the second day in two sessions, the second of those after the endoscope and setup joints were moved and the arms recalibrated. A median demonstration lasts 8.2 s for needle pickup, 28.6 s for needle throw and 10.5 s for knot tying.

Annotations. Every demonstration is labelled with its phase command; every needle-throw demonstration carries the two clicked insertion and exit points used for goal conditioning (Sec. III-E); a sub-task annotation cuts the same demonstrations into 289 clips of 11 sub-tasks with text commands.

Splits. The *first split* excludes the recalibrated session and holds out the last two demonstrations of each phase in

TABLE I
SUTUREBOT BENCHMARK, HELD-OUT TISSUE 10 (12 DEMONSTRATIONS, EVERY FRAME): PER-AXIS L1 ERROR OF THE PREDICTED 60-STEP CHUNK [MM], PSM1 / PSM2. BOLD: LOWER ENCODER PER CELL. †BEST SNAPSHOT, EVERY 10TH FRAME (SAME-SUBSET ACT NUMBERS IN THE TEXT).

Phase	ACT baseline	SurgeNet-ACT
Needle pickup	0.32 / 0.31	0.30 / 0.27
Needle throw	0.71 / 0.28	0.55 / 0.20
Knot tying	0.42 / 0.42	0.36 / 0.34
Pickup recovery	1.40 / 1.69	1.58 / 1.64
Throw recovery	1.92 / 0.98	1.65 / 0.88
Knot recovery	1.30 / 1.55	1.70 / 1.63
All phases	1.16 / 0.84	1.10 / 0.78
SurgeNet + VQ-BeT head [15]†	1.13 / 0.75	
SurgeNet + diffusion head [14]†	2.03 / 1.42	

TABLE II
SUTUREBOT BENCHMARK, TISSUE 10, SCORED WITH THE SURGLAB PER-DEMONSTRATION PROTOCOL (SEC. III-G): POSITION ERROR PSM1 / PSM2, RAW AND OFFSET-CORRECTED [MM], ROTATION [DEG], FIRST 20 STEPS; ALL 12 DEMONSTRATIONS AND THE 6 CORE-PHASE ONES. BOLD: LOWER PER COLUMN.

Policy	Demos	Raw [mm]	Offset-corr. [mm]	Rot. [deg]
ACT (EfficientNet-B3)	all	1.86 / 1.55	1.36 / 1.03	3.6
SurgeNet-ACT	all	1.74 / 1.36	1.35 / 1.01	3.7
ACT (EfficientNet-B3)	core	0.95 / 0.66	0.84 / 0.54	2.0
SurgeNet-ACT	core	0.81 / 0.57	0.75 / 0.49	1.8

recording order, one from the training session and one from the unseen second-day session (75 training, 6 validation). The *final split* uses all 84 demonstrations and holds out the two second-day demonstrations of each phase, so validation consists entirely of unseen sessions and training is the 78 first-day demonstrations. Actions are the setpoints, 20 numbers per step (Sec. III-E); chunks of 100, 60 and 30 steps are used.

Against SutureBot. SurgLab is 22 times smaller but records three sessions, four views, camera-frame kinematics and measured as well as commanded poses: the case of a laboratory reproducing the pipeline on its own robot. The dataset and splits will be released at [DATASET URL]; the training code, evaluation scripts and deployment wrapper at [CODE URL].

C. Policies and Baselines

All policies share the ACT head (Sec. III-F). *ACT baseline*: the SutureBot recipe with its EfficientNet-B3 encoder on SutureBot and a ResNet-18 encoder on SurgLab, absolute actions. *SurgeNet-ACT*: the same head on the frozen SurgeNetXL encoder, absolute actions; on SurgLab also with LoRA, with the last two ViT blocks fine-tuned, and initialised from a head pre-trained on SutureBot. + *relative actions*: tool-frame targets (1); + *history*: the temporal context of Sec. III-E. *Heads*: VQ-BeT and diffusion on SutureBot, VQ-BeT with and without history on SurgLab.

D. Results on the SutureBot Benchmark

Encoder. The frozen SurgeNetXL encoder beats the benchmark’s own EfficientNet-B3 encoder on the held-out tissue

(Table I). It lowers position error on all three core phases, by 23 percent on needle throw and 15 percent on knot tying, and on the whole tissue, 1.10 / 0.78 against 1.16 / 0.84 mm. It loses on two of the three recovery phases. Every row is a single run, so differences below about 0.05 mm are not a ranking.

Policy head. The CVAE head of ACT ties VQ-BeT and beats diffusion on offline L1. On the same frozen features and the same every-10th-frame subset, the CVAE head scores 1.11 / 0.79 mm, the best VQ-BeT snapshot 1.13 / 0.75 mm and the best diffusion snapshot 2.03 / 1.42 mm. CVAE and VQ-BeT are tied within the single-run resolution of the benchmark; we keep the CVAE head because it runs in one forward pass at 30 Hz. The diffusion head’s higher L1 must not be read as underperformance outright. The metric compares one sampled trajectory with the single demonstrated one, which penalises any head that represents a multimodal action distribution and favours by construction a head that regresses the conditional mean.

Held-out tissue under our protocol. The encoder gain also holds under the per-demonstration protocol of our robot (Table II). SurgeNet-ACT is lower on both arms at the executed horizon, over all twelve demonstrations and over the six core-phase ones; offset-corrected errors stay close to the raw ones and rotation error is 2 to 4 degrees, since the held-out tissue changes the phantom, not the recording setup. Both policies start every chunk on the current pose and differ only in the shape of the motion, as Fig. 2 shows on a held-out needle-throw demonstration.

E. Results on the SurgLab Benchmark

Encoder and adaptation. On our robot the frozen encoder lowers error again and adaptation adds nothing (Table III, first split). Every SurgeNet-based policy has lower offset-corrected error than the ResNet-18 baseline, 3.1 to 3.6 mm against 4.5 / 3.8 mm, whatever the recipe or seed. LoRA, fine-tuning the last two blocks, and initialising from a head pre-trained on SutureBot all land within the seed spread or above it.

Cross-session error and tool-frame actions. The raw error of an absolute-action policy on a new session is dominated by the calibration offset, and tool-frame actions remove it (Tables IV, III and V). On the unseen session of the first split the constant offset of the absolute policy reaches 5.5 mm and its rotation error 29 degrees, against 2 to 3 mm and 15 to 20 degrees on the training session; across all absolute models the unseen-session offsets range from 2 to 14 mm. Relative actions cut the raw error by a factor of four and the rotation error by a factor of seven, with a three-seed spread below 0.1 mm and 0.2 degrees; fine-tuning the backbone on top changes nothing. Table V shows a per-demonstration offset below 1 mm on every validation demonstration and 3 degrees of rotation error at the executed horizon. It also shows the limit of the protocol: over a full 100-step chunk no policy beats holding the current pose.

Temporal context and head. Temporal context gives a small gain, to be read against the seed spread, and the CVAE head beats VQ-BeT on our data (Table VI, 60-step chunks).

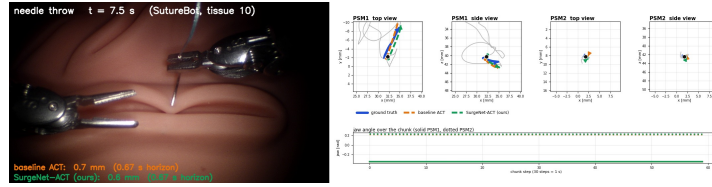


Fig. 2. SutureBot Benchmark, held-out tissue 10, needle throw: demonstrated chunk (blue), baseline ACT (orange) and SurgeNet-ACT (green) in views zoomed to ± 8 mm around each arm’s current pose (black dot), with the jaw angle over the 60-step chunk. Both policies start on the current pose, since the benchmark has one calibration; only the shape of the motion differs.

TABLE III

SURGLAB BENCHMARK, FIRST SPLIT: ENCODER, ADAPTATION AND ACTION REPRESENTATION. POSITION ERROR OVER THE FIRST 20 STEPS, PSM1 / PSM2, RAW AND OFFSET-CORRECTED [MM], ROTATION [DEG]; MEAN $\pm$ STD OVER THREE SEEDS WHERE GIVEN. BOLD: BEST PER COLUMN.

Policy	Encoder / actions	Raw [mm]	Offset-corrected [mm]	Rot. [deg]
ACT (SutureBot recipe) [4]	ResNet-18, absolute	6.74 / 5.05	4.45 / 3.84	23.5
SurgeNet-ACT, original augmentation	frozen SurgeNetXL, absolute	4.15 / 3.87	3.06 / 3.20	20.0
SurgeNet-ACT (3 seeds)	frozen SurgeNetXL, absolute	4.61 ± 0.20 / 4.41 ± 0.33	3.22 ± 0.28 / 3.29 ± 0.19	21.1 ± 0.6
+ LoRA, last 4 blocks	adapted, absolute	5.13 / 6.13	3.42 / 3.39	22.8
+ last 2 blocks fine-tuned	adapted, absolute	4.80 / 4.66	3.16 / 3.16	20.9
+ SutureBot pre-training	adapted, absolute	6.81 / 7.97	3.64 / 3.60	23.7
SurgeNet-ACT + relative actions (3 seeds)	frozen SurgeNetXL, tool-frame	1.06 ± 0.04 / 1.07 ± 0.05	1.04 / 1.03	3.1 ± 0.2
+ last 2 blocks fine-tuned	adapted, tool-frame	1.03 / 1.08	1.04 / 1.06	3.0

TABLE IV

SURGLAB BENCHMARK: PER-SESSION BREAKDOWN OF ONE ABSOLUTE-ACTION POLICY (FROZEN SURGENETXL, SEED 0, 20-STEP HORIZON). TRAINING DATA COMES FROM SESSION 08-27 ONLY. OFFSET IS THE NORM OF THE PER-DEMONSTRATION MEAN ERROR VECTOR.

Phase	Session	Raw [mm]	Offset [mm]	Corrected [mm]	Rot. [deg]
Pickup	08-27 seen	4.50 / 3.99	1.8 / 1.7	4.09 / 3.72	19.8
Pickup	08-28 unseen	3.75 / 4.26	3.5 / 3.3	1.19 / 2.58	20.4
Throw	08-27 seen	3.81 / 3.17	2.2 / 2.2	3.19 / 2.37	15.4
Throw	08-28 unseen	4.19 / 4.59	1.9 / 2.5	3.63 / 3.99	26.0
Knot	08-27 seen	3.57 / 4.74	3.1 / 3.5	1.92 / 3.25	16.4
Knot	08-28 unseen	6.45 / 6.53	5.5 / 5.4	4.15 / 4.11	28.9

TABLE V

SURGLAB BENCHMARK: RELATIVE ACTIONS AT TWO HORIZONS (THREE SEEDS), COMPARED WITH HOLDING THE CURRENT POSE; THE ABSOLUTE ROW IS THE ORIGINAL-AUGMENTATION SURGENET-ACT OF TABLE III.

Horizon	Hold	Raw [mm]	Corr. [mm]	Rot. [deg]
20 steps (0.67 s)	0.98	1.06 ± 0.04 / 1.07 ± 0.05	1.04 / 1.03	3.1 ± 0.2
100 steps (3.3 s)	3.03	3.50 ± 0.12 / 3.22 ± 0.15	3.34 / 2.85	11.0 ± 0.3
Absolute, 100 steps	–	5.20 / 4.92	4.12 / 4.20	24.1

With the CVAE head, history lowers error at both horizons and gives the only policy below the hold-pose reference of Sec. III-G at 20 and 60 steps; over three 3000-epoch seeds the gain is about 0.1 mm, two to four times the seed standard deviation. Longer history does not help, past poses alone recover most of the gain, and the VQ-BeT head is worse with and without history, unlike on SutureBot. All runs select their best checkpoint before epoch 240 and fully converged weights return to the hold-pose floor: early stopping on deployment-mode validation is part of the recipe.

Final policy on two unseen sessions. The final configuration, 30-step tool-frame chunks with history, keeps its accuracy

when every validation demonstration comes from a session the policy never saw, one of them recorded after the endoscope was moved (Table VII, final split). All six history seeds beat both no-history controls at 30 steps and five of six beat the hold-pose reference of Sec. III-G at 20 steps. The best seed reaches 0.78 / 0.79 mm and 3.0 degrees at the executed horizon; the six-seed mean of 0.95 / 0.87 mm says how much of that is seed luck. The weighted loss with a direction term produced both the best and the worst seed, so its effect lies inside seed noise.

Why offline error saturates. The hold-pose floor arises because every policy predicts the direction of fine motion weakly (Table VIII). The relative policy predicts about 80 percent of the true motion magnitude, so it has not collapsed to stillness. The direction of its motion, however, agrees with the held-out trajectories only weakly, and absolute policies are the same or worse. The validation demonstrations move about 3 mm per chunk, close to teleoperation jitter. At that scale the offline metric stops ranking policies; fine motion is left to closed-loop success.

Qualitative. Fig. 3 shows the baseline ACT and the deployed SurgeNet-ACT policy frame by frame on a held-out demonstration. The baseline chunks start 3 to 10 mm from the current pose and stay there for the whole demonstration; ours start on the measured pose by construction and stay within about a millimetre, rising briefly at the fastest motions. Table IX gives the mean error per demonstration for both unseen sessions: on session B, recorded after the endoscope was moved, the baseline error is 28 to 37 mm while ours is unchanged at 0.6 to 1.1 mm.

F. Cross-Benchmark Analysis

The frozen-encoder gain transfers: it holds on both benchmarks, on two robots, two camera sets and a 25-fold difference

TABLE VI

SURGLAB BENCHMARK, 60-STEP RELATIVE CHUNKS: TEMPORAL CONTEXT AND HEAD. POSITION ERROR PSM1 / PSM2 [MM] AND ROTATION [DEG] AT 20 AND 60 STEPS; TOP BLOCK 300-EPOCH SEEDS, BOTTOM 3000-EPOCH RUNS. BOLD: BEST PER COLUMN. HOLD REFERENCE 0.98 / 2.21 MM.

Head	History	20 steps [mm]	Rot. [deg]	60 steps [mm]	Rot. [deg]
ACT (CVAE)	no	0.99 / 1.00	3.2	2.31 / 2.18	7.5
ACT (CVAE)	yes	0.91 / 0.89	2.8	2.10 / 1.96	6.9
VQ-BeT [15]	no	1.06 / 0.96	3.0	2.48 / 2.08	8.0
VQ-BeT [15]	yes	1.01 / 0.97	2.9	2.36 / 2.24	7.5
ACT + history, 3000 epochs (3 seeds)	yes	$0.95 \pm 0.02 / 0.92 \pm 0.04$	2.9 ± 0.2	$2.16 \pm 0.06 / 2.04 \pm 0.03$	7.0
ACT, 3000 epochs	no	1.04 / 1.00	2.8	2.31 / 2.20	7.1
ACT + 16-pose history (2 seeds)	yes	0.97 / 0.94	2.8	2.26 / 2.10	7.1
ACT + poses only, no frames (2 seeds)	yes	0.97 / 0.97	3.3	2.24 / 2.16	7.2

TABLE VII

SURGLAB BENCHMARK, FINAL SPLIT (TWO UNSEEN SESSIONS), 30-STEP RELATIVE CHUNKS WITH HISTORY, BEST EPOCH IN PARENTHESES: POSITION ERROR PSM1 / PSM2 [MM], ROTATION [DEG] AT 20 AND 30 STEPS. BOLD: BEST RUN PER COLUMN. HOLD REFERENCE 0.88 / 1.25 MM.

Policy	Loss	20 steps [mm]	Rot. [deg]	30 steps [mm]	Rot. [deg]
ACT + history, seed 0 (124)	L1	0.99 / 0.96	3.2	1.44 / 1.35	4.5
ACT + history, seed 1 (17)	L1	0.89 / 0.84	3.0	1.20 / 1.16	4.2
ACT + history, seed 2 (23)	L1	0.81 / 0.80	3.2	1.14 / 1.10	4.4
ACT + history, seed 0 (14)	weighted, dir.	1.22 / 0.93	3.4	1.49 / 1.24	4.5
ACT + history, seed 1 (19)	weighted, dir.	0.99 / 0.89	3.2	1.26 / 1.20	4.3
ACT + history, seed 2 (52)	weighted, dir.	0.78 / 0.79	3.0	1.07 / 1.11	4.2
ACT + history, mean of 6 seeds	both	$0.95 \pm 0.16 / 0.87 \pm 0.07$	3.2 ± 0.1	$1.27 \pm 0.16 / 1.19 \pm 0.09$	4.4 ± 0.1
ACT, no history (95)	L1	1.00 / 0.98	3.1	1.41 / 1.37	4.3
ACT, no history (30)	weighted, dir.	1.01 / 1.06	3.1	1.32 / 1.37	4.3
Best 60-step policy (Table VI) on this set	L1	1.23 / 1.04	3.5	1.73 / 1.44	4.8

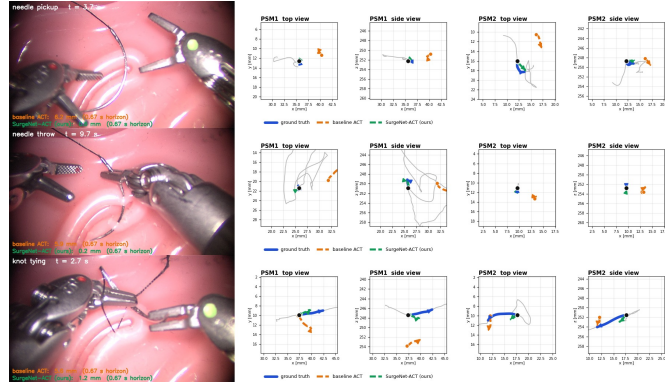


Fig. 3. SurgLab Benchmark, unseen session A, one instant per phase: demonstrated chunk (blue), baseline ACT (orange) and SurgeNet-ACT (green), zoomed to ± 8 mm around each arm's current pose (black dot; grey: demonstrated path). The baseline is displaced by the calibration offset; ours starts on the measured pose.

TABLE VIII

SURGLAB BENCHMARK: DIRECTION AGREEMENT BETWEEN PREDICTED AND DEMONSTRATED TOOL-FRAME MOTION ON THE VALIDATION DEMONSTRATIONS.

Policy	Horizon	Magnitude ratio	Cosine
Relative actions	20	0.74	0.25
Relative actions	100	0.79	0.34
Absolute, original augmentation	100	0.68	0.22
Absolute, ours	100	0.75	0.27

in data. The order of heads does not: CVAE and VQ-BeT tie on SutureBot but VQ-BeT loses on SurgLab at every horizon, which we attribute to 25 times fewer demonstrations. The error

scale does not transfer either (Table II against Table III): the gap is the data and the sessions, not the policy.

G. Ablations

Head hyperparameters are not the bottleneck. A nine-run sweep of the absolute-action SurgeNet-ACT on the first split over learning rate, KL weight, dropout, camera embedding, rotation loss and EMA moved validation L1 by at most 0.012 and left rotation error at 23 degrees.

V. DISCUSSION

SurgeNet-ACT works because each stage removes one thing that 78 demonstrations must otherwise teach. The frozen SurgeNetXL encoder brings the appearance of needle, jaw and

TABLE IX

SURGLAB BENCHMARK: MEAN POSITION ERROR PER HELD-OUT DEMONSTRATION (BOTH ARMS, 0.67 S HORIZON, MM), BASELINE ACT VS. DEPLOYED SURGENET-ACT. SESSION B FOLLOWED A MOVE OF THE ENDOSCOPE AND A RECALIBRATION.

Phase	Session	ACT	Ours
needle pickup	A	5.47	0.68
needle pickup	B (recalib.)	36.51	1.05
needle throw	A	6.90	0.59
needle throw	B (recalib.)	30.96	0.59
knot tying	A	8.50	0.89
knot tying	B (recalib.)	28.02	0.99

tissue from 4.7 million surgical frames, and the adaptation rows of Table III stay inside the seed spread. What the backbone buys is features, not control. The frozen encoder wins on every core phase of SutureBot’s held-out tissue (Fig. 2) but loses on two of the three perturbed recovery phases (Table I); on SurgLab absolute targets keep a rotation error near 20 degrees (Table III) that tool-frame chunks remove.

Only the two benchmarks together separate the encoder from the data. SutureBot shows that ACT’s CVAE head ties VQ-BeT on offline L1 and that the encoder gain survives our per-demonstration protocol (Tables I and II); SurgLab shows that the gain survives 25 times fewer demonstrations while VQ-BeT loses at every horizon (Tables III and VI). Head rankings do not transfer between benchmarks; encoder rankings do.

Action chunking helps at the executed horizon and hurts over the full chunk. At 20 steps the final 30-step policy beats the hold-pose reference on five of six seeds (Table VII); over a 100-step chunk no policy beats holding the current pose (Table V). The direction diagnostic (Table VIII) explains why: demonstrated motion is about 3 mm per chunk and every policy predicts its direction weakly, so longer chunks accumulate direction error faster than foresight (Fig. 3). The diffusion head’s higher offline L1 (Table I) reflects a metric that scores one sample against one demonstration, not worse control.

The SurgLab Benchmark makes the cross-session question testable by others. Its three sessions, measured and commanded poses, four views and per-demonstration protocol (Sec. IV-B) let any dVRK laboratory add sessions recorded on other days to Tables III and VII.

Limitations. All evidence is offline, because SutureBot’s robot is not ours; closed-loop trials on our dVRK must report success rate. Six validation demonstrations from two sessions and several single-seed rows bound the resolution; each session costs a day of recording. No frozen natural-image ViT of the same size was run, so the gain is not attributed to surgical pre-training alone; the missing control is a natural-image DINOv3 ViT-B/16.

VI. CONCLUSION

SurgeNet-ACT pairs a frozen surgical foundation encoder with an ACT policy that predicts tool-frame action chunks.

Trained on 78 dVRK suturing demonstrations, it reaches 0.78 / 0.79 mm and 3.0 degrees on two unseen sessions. The SurgLab Benchmark and the code are released at [URL]. All evidence is offline; closed-loop trials on the dVRK are the next step.

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